JOURNAL OF GEOPHYSICAL RESEARCH, VOL. 89, NO. D5, PAGES 7231–7237, AUGUST 20, 1984

Land Surface Temperature Measurements From the Split Window Channels of the NOAA 7 Advanced Very High Resolution Radiometer

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The Advanced Very High Resolution Radiometer (AVHRR) on the NOAA 7 satellite acquires 1-km spatial resolution data in "split window" channels at 10.8 and 11.9 μm. Data from these spectral channels may be used to estimate surface temperature and the atmospheric correction to radiation from the earth's surface. Analysis of a data set from July 1981 shows that (1) there is satisfactory agreement between the equation resulting from radiative transfer theory and the atmospheric correction algorithm as obtained by analysis of an area of incipient cloud street formation; (2) agreement is also satisfactory between this algorithm and the statistically derived NOAA algorithm used to obtain sea surface temperatures from the satellite data (However, the comparison assumes the NOAA algorithm is valid outside its range of derivation.); (3) in areas of cloud street formation, variations of atmospheric moisture produce radiance temperature differences of order 2–3°C, which if neglected would cause errors in the derivation of surface thermal characteristics. This meteorological variation over distances of 5–10 km would not be inferred from conventional radiosonde measurements or from lower-resolution satellite soundings.

INTRODUCTION

Since the late 1960's, satellite thermal infrared measurements have been utilized to provide estimates of sea surface temperature [*Curtis and Rao*, 1969; *La Violette and Chabot*, 1968; *Rao*, 1968]. With the passage of time, improved satellite sensors have produced higher and higher quality data, culminating in the present Advanced Very High Resolution Radiometer (AVHRR), which is carried on NOAA operational satellites [*Kidwell*, 1981]. Although the 1.1-km resolution at nadir of this instrument is better than required to resolve oceanographic temperature structure, this resolution is very desirable for observing between clouds, which are usually present over the oceans and which can produce low estimates of sea surface temperature. An extensive literature has developed concerning the theory and application of infrared remote sensing to obtain sea surface temperatures [*Prabhakara et al.*, 1974; *Walton*, 1980; *McClain et al.*, 1983].

For the estimation of land surface temperatures the situation is quite different; the 1.1-km field of view is not adequate to resolve many surface features, as may be illustrated with visible and near-infrared data from Landsat. Because the temperature of bare soil may vary by several degrees over distances of 10's of meters [*Vauclin et al.*, 1982] and temperature variations between cropped and fallow areas can be much greater, it is not feasible to seek accurate ground truth verification for AVHRR temperature data in agricultural regions where surface temperature data may prove useful. In addition the physical interpretation of land surface temperatures is complicated because land temperatures cycle through a considerable temperature range each day in response to varying solar input and radiative heat loss at night. On the other hand, highly accurate values of surface temperature are not required for some purposes [*Rosema et al.*, 1978; *Carlson et al.*, 1981; *Price*, 1982]. Since the AVHRR carries visible (0.6–0.7 μm) and near-infrared (0.8–1.1 μm) channels as well as three thermal infrared channels, the data represent a potentially valuable source of information concerning vegetation and moisture conditions in inhabited areas. The present work seeks to extend this capability through more accurate estimates of land surface temperatures. Several differences of approach result from the land versus ocean application. Measurements of ocean surface temperatures have an implied statistical variability because of the uncertainty in locating precisely a given latitude and longitude in the relatively featureless satellite imagery over an ocean [*McClain*, 1983] and because a radiance derived for the surface skin temperature does not equate directly to a measurement by a sensor suspended 1–2 m beneath the ocean surface. For the land surface, recognizable features make it possible, at least in principle, to locate the satellite-derived temperatures and to relate them to local conditions. In addition the natural variability of land surface types over tens to hundreds of square kilometers suggests the analysis of spatial variability of surface temperatures may be useful, even if highly accurate values, e.g., ±1°C, are not feasible.

Thus the present work complements those of *Deschamps and Phulpin* [1980] and *McClain et al.* [1983], to which we refer the reader for both background and application of some of the material following. Section 2 reviews the theory of radiation measurements in the 10–13 μm region of the spectrum, developing a formulation suitable for high spatial resolution data sets over extensive agricultural areas. In section 3, results from a satellite data set over the south central United States are compared with those from radiosonde data, and possible error sources are identified. Incipient formation of cloud streets, i.e., parallel rows of cumulus clouds, permits estimation of the relative magnitudes of atmospheric absorption in the 10.9 and 11.8-μm channels while illustrating the extent to which atmospheric effects can mimic surface temperature variations. The implications for remote sensing of land surface temperatures and effects of surface emissivity variation are discussed in section 4.

2. *Radiation Theory for Atmospheric Correction by Differential Absorption*

The formulation presented here follows that of *Deschamps and Phulpin* [1980] with emphasis on the form and magnitude of the atmospheric transmission coefficients. The equation of radiative transfer [*Chandrasekhar*, 1960] is written

$$\frac{dI_\lambda}{ds} = -k_\lambda U(I_\lambda + B_\lambda)$$

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Paper number 4D0556.

where I_λ is the intensity of radiation at wavelength λ (μm) passing through an absorber U (kg $-$ m^{-3}) and having an absorption coefficient k_λ (m^2 $-$ kg^{-1}). Here $B_\lambda(T(s))$ is the blackbody radiance as given by the Planck function. Following the change of variables from distance s to optical depth τ_λ, i.e., $d\tau_\lambda = k_\lambda U ds$, a formal integration yields the satellite-observed radiation $I_\lambda(T_{BB\lambda})$ at the top of the atmosphere

$$I_\lambda(T_{BB\lambda}) = B_\lambda(T_0) - \int_0^{\tau_\lambda} d\tau_\lambda' e^{-\tau_\lambda'}[B_\lambda(T_0) - B_\lambda(T(\tau_\lambda - \tau_\lambda'))] \quad (1)$$

Here the surface emission and the satellite-observed radiance are identified with that from equivalent blackbodies, i.e., emissivity equals 1, at temperatures T_0, $T_{BB\lambda}$, and the atmospheric emission is characterized by temperature T at optical depth $\tau_\lambda - \tau_\lambda'$. The effect of nonunit emissivity is discussed in sections 3 and 4. For simplicity it is desirable to expand the Planck functions about T_0, keeping only constant and linear terms, as is valid for temperatures near the surface radiance temperature T_0. This is valid when the absorber (water vapor) is concentrated in the lower atmosphere, where temperatures are generally within 10–20°C of the surface temperature. Following this approximation, equation (1) may be written

$$T_{BB\lambda} = T_0 - \int_0^{\tau_\lambda} d\tau_\lambda' e^{-\tau_\lambda'}[T_0 - T(\tau_\lambda - \tau_\lambda')] \quad (2)$$

where the second term represents the influence of the atmosphere. Equation (2) correctly states that the satellite-observed temperature approaches the ground radiance temperature as the optical depth τ_λ becomes smaller.

In the 10–13 μm window, aerosol scattering may be neglected, except for hazy conditions [*McClatchey et al.*, 1971], while clouds render this treatment inapplicable. The dominant effect results from absorption and reemission of radiation by water vapor. For weak absorption $\tau_\lambda \ll 1$, equation (2) may be expanded, keeping only first-order terms in τ_λ:

$$T_{BB\lambda} = T_0(1 - \tau_\lambda) + \int_0^{\tau_\lambda} d\tau_\lambda' T(\tau_\lambda - \tau_\lambda')$$

or

$$T_{BB\lambda} = T_0(1 - \tau_\lambda) + \int_0^{U_{path}} dU k_\lambda T(U) \quad (3)$$

where $d\tau_\lambda = k_\lambda dU$.

This is similar to equation (8) of *Deschamps and Phulpin* [1980]. Thus the temperature correction is an absorber-weighted (dU) average of the difference between surface temperature and the local temperature of the absorber. Note that the assumption $\tau_\lambda \ll 1$ is sometimes not strongly satisfied, implying second-order corrections to (3).

At this point the reason for obtaining two measurements in the 10–13 μm atmospheric window may be illustrated. If k_λ may be expressed as the product of a wavelength-dependent scaling factor $C(\lambda)$ and some function of atmospheric conditions $f(U, T, p)$, i.e., $k_1 = C(\lambda_1)f$, $k_2 = C(\lambda_2)f$, then a solution for T_0 in terms of $T_{BB\lambda_1}$ and $T_{BB\lambda_2}$ is elementary. This basic assumption is commonly used in the reduction of laboratory data to transmission coefficients in this spectral interval [*Kneizys et al.*, 1980]. After defining $U_T = \int dU f(U, T, p)[T_0 - T(U)]$, equation (3) for the two wavelengths has the form

$$T_{BB\lambda_1} = T_0 - C(\lambda_1)U_T$$

$$T_{BB\lambda_2} = T_0 - C(\lambda_2)U_T$$

These equations may be solved for the surface radiation temperature

$$T_0 = T_{BB\lambda_1} + \frac{1}{[C(\lambda_2)/C(\lambda_1) - 1]}(T_{BB\lambda_1} - T_{BB\lambda_2}) \quad (4)$$

Thus the explicit dependence through U_T on atmospheric conditions drops out, and the solution depends on the ratio $R = C(\lambda_2)/C(\lambda_1)$ but not on the magnitudes of k_1 or k_2 separately.

Although the absorption coefficient of water vapor in the 10–13 μm window has been studied extensively, uncertainty about the magnitude and wavelength dependence remains. *Imbault* [1979] has reviewed the literature and summarized the experimental findings. The water vapor absorption coefficient is generally written in the form $k_\lambda = K_{1\lambda}p + K_{2\lambda}e$, where p is atmospheric pressure, e is water vapor pressure (atm), and the vapor pressure term is dominant in the lower atmosphere. Evidently, the above analysis requires that the ratio K_2/K_1 be independent of wavelength. More significant is the variability in experimental determinations of K_1 and K_2 (Figures 1 and 2). It is evident that both the magnitude and the wavelength dependence of k_λ are not well established. Since the numerical value of k_λ provides the basis for radiative transfer computations, it appears that additional experimental evidence is desirable. In the next section an estimate is obtained for $C(\lambda_2)/C(\lambda_1)$, and the result is compared with the NOAA sea surface temperature algorithm and with that predicted by the expression of *Kneizys et al.* [1980] for k_λ:

$$k_\lambda = [4.18 + 5578 \exp(-78.7/\lambda)] \cdot \{\exp[1800(1/T - 1/296)]\}(p + 0.002e)$$

This assumption is consistent with the derivation of equation (4) because K_2/K_1 is independent of wavelength.

3. Application to a Satellite Data Set

For identification of agriculturally related parameters an AVHRR data set was selected, representing the central United States on July 20, 1981 (Figure 3). Radiosonde data required for radiative transfer computations were obtained for usual morning and evening observation times: 12Z on July 20 and 00Z on July 21. Although this timing misses the midafternoon heating of the lower atmosphere, it should permit a reasonable estimate of the atmospheric influence on the satellite thermal infrared data. Table 1 presents the surface air temperature taken in conjunction with the radiosonde observations and the atmospheric correction to the 10.9 and 11.8 μm AVHRR observations as computed from theory for a surface temperature assumed 10°C above the measured air temperature. This assumption regarding surface temperature is for purposes of illustration, as the atmospheric correction is a function of both the surface temperature and the atmospheric state [*Price*, 1983]. The variability of $R = C(\lambda_2)/C(\lambda_1)$ in column 6 demonstrates that the assumptions leading to equation (4) are not rigorously satisfied.

For the July 20 data set the satellite estimates of land surface temperatures were computed by assuming (from Table 1) the mean value $R = 1.36$ and the value for emissivity $\varepsilon = 0.96$ [*Taylor*, 1979]. The derived surface temperatures are of the order of 30° to 55°C, which is well within the expected range, considering time of day; variability of surface conditions, such as emissivity; surface evaporation, etc. It is noteworthy that satellite-derived surface temperatures are well above midafternoon air temperatures in the western areas, suggesting that these areas are dry at the surface and have low evapo-

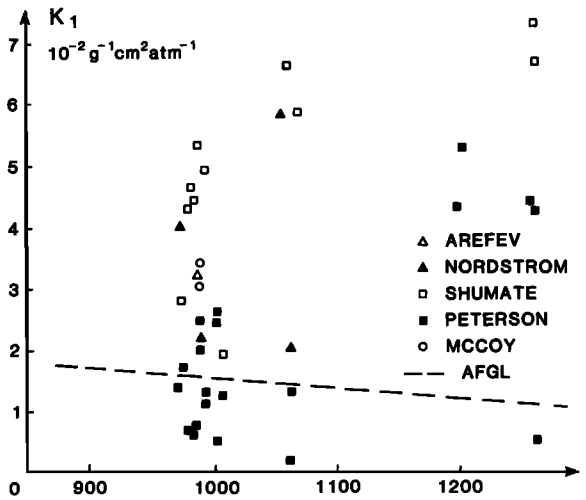

Fig. 1. The K_1 coefficient for water vapor absorption [from *Imbault*, 1979]. In practice the contribution from this term is less important than the K_2 term. AVHRR channel 4 = 927 cm^{-1}; channel 5 = 840 cm^{-1}.

transpiration rates. In such cases, surface temperatures can exceed 70°C, showing that comparison with air temperature is not always a good indicator of surface temperature.

3a. *Evaluation of the Ratio R From an Area of Cloud Street Formation*

Inspection of the image data reveals the presence of well-defined cloud streets in the south Texas area. The orientation agrees well with surface observations of a weak southwesterly flow produced by a surface high-pressure center over the Gulf of Mexico [*National Oceanic and Atmospheric Administration*, 1981]. Detailed examination of visible and thermal infrared data shows that, as expected, the bright clouds in the visible channel data are at the middle of long streaks of relatively higher atmospheric moisture, as inferred from relatively lower temperatures in the thermal infrared channels. Unfortunately, in portraying this comparison, photographic processing yields a poor substitute for results from a digital image display system.

Because of both the importance of R in estimating surface temperature and the paucity of relevant experimental data, an estimate of R was obtained by using data in an area showing atmospheric variability. A small area perpendicular to incipient cloud streets in a region where visible clouds had not yet formed (Figure 4) was selected. Figure 5 illustrates the brightness temperature variation in the 10.8 and 11.9 μm channels as averaged over 10 consecutive data lines crossing the cloud street pattern. It is assumed that the surface temperature in this area is uniform because the area is generally devoid of surface features at this spatial scale. This was verified through examination of the coincident visible and near-infrared data (AVHRR channels 1 and 2). Given this assumption, the observed variations of brightness temperatures are due to atmospheric nonuniformity. Thus it is possible to compute R, i.e., given

$$T_1(x) = T_0 - C(\lambda_1)U_T(x)$$

$$T_2(x) = T_0 - C(\lambda_2)U_T(x)$$

one may compute the spatially averaged quantities according to

$$\langle\psi\rangle = \frac{1}{L}\int_0^L dx\psi(x)$$

$$\langle T_1\rangle = \langle T_0\rangle - C(\lambda_1)\langle U_T\rangle$$

$$\langle T_2\rangle = \langle T_0\rangle - C(\lambda_2)\langle U_T\rangle$$

and the variation about the average

$$\delta\psi \equiv \psi(x) - \langle\psi\rangle$$

$$\delta T_1 = -C(\lambda_1)\,\delta U_T$$

$$\delta T_2 = -C(\lambda_2)\,\delta U_T$$

from which $R = C(\lambda_2)/C(\lambda_1) = \delta T_2/\delta T_1$. Since the ratio is susceptible to instrument noise and to nonuniformity of surface brightness temperature, it is preferable to compute R from averaged quantities, i.e.,

$$R = \langle T_1\delta T_2\rangle/\langle\delta T_1{}^2\rangle$$

For the data illustrated in Figure 4 this yields the result $R = 1.30$, so that

$$T_s = T_{10.8} + 3.33(T_{10.8} - T_{11.9}) \quad (5)$$

Because this R value is derived from variations about a mean rather than from absolute values of temperature, it is relatively insensitive to errors in calibration of the satellite, to spectral differences in surface emissivity between the two AVHRR channels, and to the actual value of surface temperature, provided only that the surface represents a uniform background on which atmospheric variation is superposed. This value of R is slightly smaller than the 1.36 predicted from radiative transfer theory and the NOAA values of section 3*b*.

3b. *Comparison With the NOAA Algorithm*

Sea surface temperature values are derived operationally through reduction of AVHRR data by NOAA's National Environmental Satellite Data and Information Service (NESDIS). The algorithm is of interest because it will soon be used to derive land surface temperatures as a possible input parameter in agricultural yield forecasting in the Department of Agriculture (Dan Tarpley, private communication, 1983). Several equations have been formulated, depending on the suitability of the channel 3 (3.7 μm) data for surface temper-

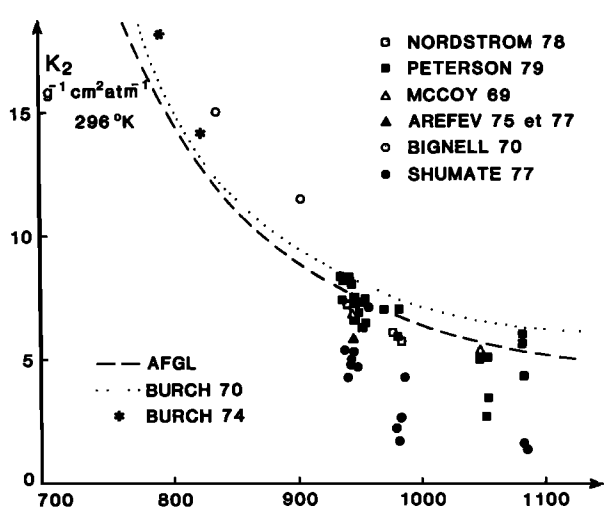

Fig. 2. The K_2 coefficient for water vapor absorption [from *Imbault*, 1979]. Note that the variation of K_2 with wavelength is not well established in the wavelength range of the AVHRR.

Fig. 3. AVHRR channel 1 image of the south central United States on July 20, 1981. Data were acquired at 20Z by NOAA 7.

ature estimation. Channel 3 data includes the effect of reflected solar radiation during the day and is strongly influenced by localized high-temperature sources at night. Of course, these sources, such as power plants, gas flares, etc., do not occur in the oceans.

Thus for comparison purposes, only the NOAA equations that use the 10.8 and 11.9 μm data are examined. These equations were developed by statistical fitting based on radiative transfer calculations with surface temperature measurements by drifting buoys [*McClain et al.*, 1983]. Different equations are obtained by day and night, possibly because the difference between subsurface buoy temperatures and the satellite-observed skin temperatures varies between day and night.

These equations are given by

$$T_{SST} = 1.035\ T_{10.8} + 3.046(T_{10.8} - T_{11.9}) - 283.93 \text{ (day)}$$

$$T_{SST} = 1.076\ T_{10.8} + 3.168(T_{10.8} - T_{11.9}) - 296.23 \text{ (night)}$$

TABLE 1. Estimation of R From Radiative Transfer Theory

Location	Date	Air Temperature, 2 m	$T_s - T_{10.8}$	$T_s - T_{11.9}$	R
Brownsville, TX	July 20	297.5	5.5	7.4	1.31
	July 21	305.9	8.6	11.6	1.34
Monett, MO	July 20	297.0	4.4	6.0	1.34
	July 21	305.6	5.8	8.1	1.37
Amarillo, TX	July 20	294.8	2.0	2.9	1.40
	July 21	308.9	7.4	10.4	1.39
Topeka, KA	July 20	297.0	3.5	4.9	1.37
	July 21	304.8	5.6	7.8	1.38

where T_{SST} is in degrees Celsius and the radiance temperatures are in degrees Kelvin. These equations may be recast in more convenient form by converting all temperatures to Celsius:

$$T_s = T_{10.8} + 3.046(T_{10.8} - T_{11.9}) + 0.035(T_{10.8} - 34.9) \quad (6)$$

$$T_s = T_{10.8} + 3.168(T_{10.8} - T_{11.9}) + 0.076(T_{10.8} - 30.5) \quad (7)$$

For temperatures of order 20–40°C the terms at the right of (6) and (7) are small, so that it is possible to estimate R, i.e., $R_{day} = 1 + 1/3.046 = 1.33$ and $R_{night} = 1 + 1/3.168 = 1.32$. These values agree well with radiative transfer theory $R = 1.36$ and with the value $R = 1.30$ as obtained in section 3*a*.

Obviously, the results are sensitive to possible errors in calibration of the satellite data. The effect of small errors in the satellite data may be evaluated from (4). Thus

$$\frac{dT_s}{dT_{10.8}} = 1 + \frac{1}{R-1} \sim 4 \qquad \frac{dT_s}{dT_{11.9}} = \frac{-1}{R-1} \sim -3.3$$

so that calibration errors are magnified in the estimate of surface temperature.

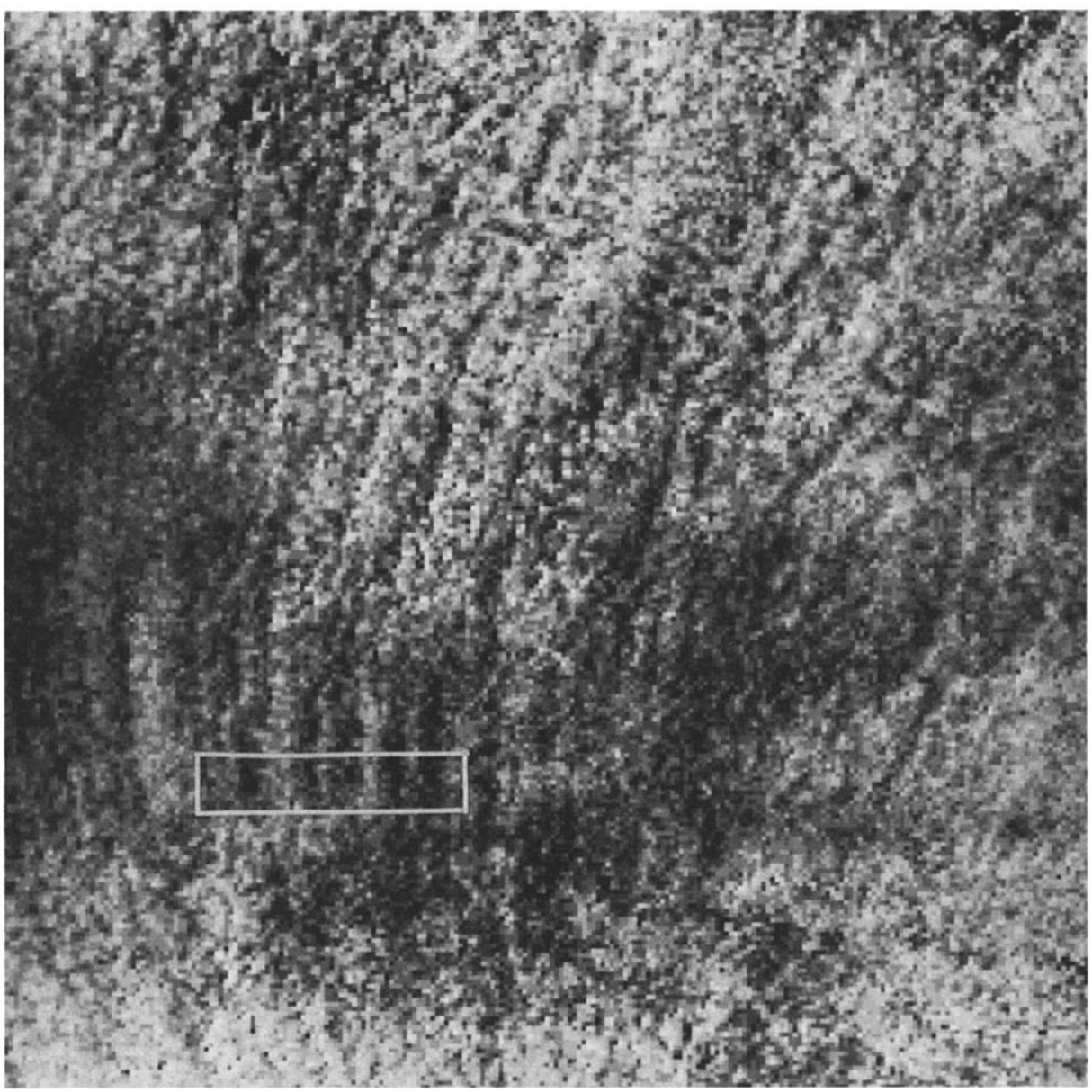

Fig. 4. An area of cloud streets in the lower left portion of Figure 3. The area is approximately (250 km)^2. The quantity displayed is the difference between AVHRR channels 4 and 5, with darker areas having higher atmospheric moisture and brighter areas representing lower atmospheric moisture. This difference image has been greatly enhanced, leaving the result very grainy.

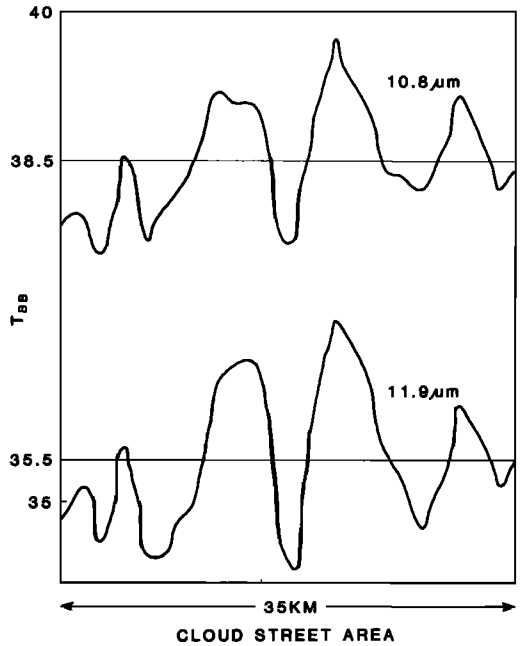

Fig. 5. Radiance temperature values corresponding to the small boxed area in Figure 4.

4. Summary and Conclusions

An initial comparison of data from the 10.8 and 11.9 μm channels of the AVHRR with radiative transfer theory shows reasonable agreement of the derived ratio of absorption strength $k_{11.9}/k_{10.8} = 1.30$ and theory (1.36). For agricultural applications an uncertainty of several degrees is unimportant for observations during the growing season; the July 20 data set shows more than a 20°C range of land surface temperatures, corresponding to a substantial range of near-surface moisture and surface evaporation. A more serious limitation of the data arises from a maximum allowable value of the thermal infrared channels at approximately 320 K radiance temperature. In the present data set, saturation indicates some areas exceed this peak temperature, so that discrimination of surface properties in these areas is not possible. Although increasing this maximum temperature limit would cause only a small decrease in sensitivity at lower temperatures—of the order of hundredths of a degree Celsius—this modification would be objectionable to those carrying out oceanographic and ice mapping applications (H. Yates, private communication, 1981).

Another potential complication arises from the possible effects of emissivity variations of natural surfaces between the 10.8 and 11.9 μm channels [*Vincent et al.*, 1975]. Although experimental data are limited on this point, it appears that surfaces in agricultural areas are nearly black ($\varepsilon \sim 0.97$) in this spectral interval [*Taylor*, 1979], and a value of 0.96 may be assumed for vegetation. The effect of emissivity is accounted for by using the fact that $I_\lambda \sim \varepsilon_\lambda T^{n_\lambda}$, where $n_{10.8} \sim 4.5$, $n_{11.9} \sim 4.3$ [*Slater*, 1980, Figures 9 and 10]. Thus, for $\varepsilon \sim 1$, the ground temperature T_s is related to the radiance temperature T_λ by

$$T_s = \frac{T_\lambda}{\varepsilon_\lambda^{1/n_\lambda}} \simeq T_\lambda\left(\frac{n_\lambda + \varepsilon_\lambda - 1}{n_\lambda}\right)$$

and the surface temperature as estimated from AVHRR channels 4 and 5 may be approximated by

$$T_s = [T_{10.8} + 3.33(T_{10.8} - T_{11.9})]\left(\frac{3.5 + \varepsilon_{10.8}}{4.5}\right) + 0.75T_{11.9}(\varepsilon_{10.8} - \varepsilon_{11.9})$$

with all temperatures in Kelvin. Thus, at 300 K, an emissivity difference of 0.01 between the two channels causes a change of 2° in the surface temperature estimate.

It is useful to estimate an error budget for the determination of land surface temperatures from the AVHRR data. Errors may arise from three sources: the satellite instrumentation, the assumption of a value for R, and the lack of knowledge of surface emissivity. The following estimates appear realistic:

(1) Radiometric calibration: In view of the statistical comparison by NOAA of satellite-derived sea surface temperatures and in situ measurements, one may presume that the radiometric calibration is valid to within 1°C. The instrument noise of the AVHRR is very low (of order tenths of a degree), so the principal effect is on the temperature difference between channels 4 and 5. These random variations yield errors well under 1°C.

The value of the ratio R, since R is not a true constant but is weakly dependent on the state of the atmosphere: The values developed in this paper (1.30, 1.32, 1.33, and 1.36) suggest a preferred value of 1.33 and a variability of 0.03. The resultant error of estimation is $\delta T_E \sim [\delta R/(R-1)^2](T_{10.8} - T_{11.9}) \approx 2$°C, assuming an interchannel temperature difference of 5°C.

Surface emissivity: From *Taylor*'s [1979] data the value $\varepsilon = 0.96$ with a variability of ± 0.01 is assumed, leading to an error of order 1°C for surface temperature. Data on the wavelength dependence of vegetation and soil emissivity is not readily available, but there is no reason to suspect significant variability, i.e., $\delta\varepsilon_\lambda \sim 0.005$ implies $\delta T_E \sim 1$°C. Both values may be significantly larger in desert and mountainous regions where exposed rocks may have significant spectral patterns.

Thus the overall estimate is for 2–3°C rms error in the estimates of land surface temperature. This is small compared to satellite-observed temperature variations but substantial in the context of methods that use the difference between crop canopy temperature and air temperature to estimate crop moisture stress [*O'Toole and Hatfield*, 1983]. The field of view of the AVHRR is so large that derived surface temperatures must be considered as good or better than the present capability for verification with ground or aircraft measurements.

In summary the new five-channel AVHRR appears to represent a significant step toward routine monitoring of large-area surface temperatures in two respects: (1) It permits direct interpretation of the satellite data without the need for radiative transfer computations. (2) It permits inference of and correction for localized atmospheric variations during conditions of active cloud formation. The latter information is not available from any other source.

Acknowledgment. I am indebted to R. J. Brown and a reviewer for suggestions that have improved the manuscript. This paper is a contribution from USDA-ARS, Hydrology Laboratory, Plant Physiology Institute, Beltsville, Maryland.

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(Received January 31, 1984;
revised April 9, 1984;
accepted April 10, 1984.)